

CS305 Machine Learning - Syllabus

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Course Information

- Course code: CS305
- Credits: 3
- Semester: 2025-26 EVEN
- Prerequisites: MA201 Probability and Statistics

Course Objectives

- Understand supervised and unsupervised learning algorithms.
- Evaluate models using appropriate metrics and validation strategies.
- Apply machine learning libraries to real datasets.

Unit 1: Foundations

Unit 1 of CS305 covers types of learning, bias-variance trade-off, train, validation and test splits, feature scaling and overfitting and regularization.

Unit 2: Regression and Classification

Unit 2 of CS305 covers linear regression, logistic regression, k-nearest neighbours, naive Bayes and support vector machines.

Unit 3: Trees and Ensembles

Unit 3 of CS305 covers decision trees, information gain and Gini index, random forests, gradient boosting and feature importance.

Unit 4: Unsupervised Learning

Unit 4 of CS305 covers k-means clustering, hierarchical clustering, DBSCAN, principal component analysis and anomaly detection.

Unit 5: Neural Networks and Evaluation

Unit 5 of CS305 covers perceptron and backpropagation, activation functions, precision, recall and F1 score, ROC and AUC and cross-validation.

Course Outcomes

- CO1: Choose a suitable learning algorithm for a problem.
- CO2: Train and tune regression models.
- CO3: Build ensemble classifiers.
- CO4: Apply clustering and dimensionality reduction.
- CO5: Evaluate models with precision, recall and ROC curves.

Assessment

CS305 is assessed out of 100 marks: 40 internal marks and 60 external marks. Internal marks consist of two Continuous Assessment Tests (CAT 1 and CAT 2) of 15 marks each and assignments worth 10 marks. The end-semester examination is conducted for 100 marks and scaled to 60. Lab work carries 20 of the 40 internal marks, and the CAT weight is halved.

Textbook

Aurelien Geron, Hands-On Machine Learning with Scikit-Learn and PyTorch.